Ch2. Basics

- **Noises**: i.i.d. noise, WN(μ, σ^2), IWN(μ, σ^2), normal white noise.
- ACVF : $\gamma_{t,s} = \text{Cov}(X_t, X_s)$ ACF : $\rho_{t,s} = \text{Corr}(X_t, X_s)$. Positive semi-definite
- **Stationary**:
  - Weak: same mean and ACVF
  - Strong: same distribution after lagged
  - If variance finite, strong $\Rightarrow$ weak
- PACF : $\phi_{kk} = \text{Corr}(R_{t|t+1,\ldots,t+k-1}, R_{t+k|t+1,\ldots,t+k-1})$
  $\phi_{11} = \rho_1$, $\phi_{kk} = \frac{\det A}{\det B}$, where $A_{ij} = B_{ij} = \rho_{i-j}$, $A_{in} = \rho_i$
- **Estimation**:
  - Mean: $\bar{y} = \dfrac{\sum y_t}{n}$, unbiased. $\text{Var}(\bar{y}) = \dfrac{\gamma_0}{n} \sum_{k=1-n}^{n-1} \left(1 - \dfrac{|k|}{n}\right) \rho_k$
    If variance $\to 0$, then consistent
- ACVF : $\hat{\gamma}_k = \dfrac{1}{n} \sum (y_t - \bar{y}_n)(y_{t+k} - \bar{y}_n)$ $\hat{\hat{\gamma}}_k = \dfrac{1}{n} \sum (y_t - \bar{y}_n)(y_{t+k} - \bar{y}_n)$
  Biased. Former better, as less variance and p.s.d.
- ACF : $\hat{\rho}_k = \dfrac{\hat{\gamma}_k}{\hat{\gamma}_0}$ PACF: same formula with calculation
  For white noise: $\text{Var}(\hat{\rho}_k) = \text{Var}(\hat{\phi}_{kk}) = \sqrt{\dfrac{1}{n}}$

Ch3. ARMA

- AR(p) : $y_t = \sum \phi_k y_{t-k} + \epsilon_t$, $\phi(B) y_t = \epsilon_t$ Stationary: roots of $\phi(z)$ outside unit circle
  Average length of stochastic cycle: $T_0 = \dfrac{2\pi}{\arccos(\phi_1 / 2\sqrt{-\phi_2})}$
  MA representation: $y_t - \mu = \dfrac{\epsilon_t}{\phi(B)}$
- Estimation (Yule-Walker): $\gamma_p = \Gamma_p \phi$, $\Gamma_{p,ij} = \gamma_{i-j}$, $\hat{\sigma}^2 = \hat{\gamma}_0 - \hat{\gamma}_p' \hat{\Gamma}_p^{-1} \hat{\gamma}_p$, $\sqrt{n}(\hat{\phi} - \phi) \sim N(0, \sigma^2 \Gamma_p^{-1})$
- MA(q) : $y_t = \mu + \epsilon_t + \sum \theta_i \epsilon_{t-i} = \mu + \theta(B) \epsilon_t$
  Always w.s., invertible iff roots outside unit circle
  AR representation: $\epsilon_t = \dfrac{y_t - \mu}{\theta(B)}$
- ARMA(p, q): $\phi(B) y_t = \theta(B) \epsilon_t$
  - Weakly stationary iff roots of $\phi(z)$ outside unit circle
  - Invertible iff roots of $\theta(z)$ outside unit circle
  - MA/AR representation similar
- ARIMA: add difference in ARMA, SARIMA: add seasonal trend in ARIMA

Ch4. Forecast

- Forecast formula: $E(Y_{t+h} \mid Y_t)$
- MSE: AR(p) : $\sigma^2, (1 + \phi_1^2)\sigma^2, \ldots$ MA(q) : $\sigma^2 \sum \theta_i^2$

Ch6. Model Building

- Box-Jenkins approach: model specification, estimation, diagnostic checking
- Model specification:
  - Use ACF, PACF, EACF to select p and q
  - MSC $= \ln(\widetilde{\sigma}^2(p, q)) + c_n \psi(p, q)$

- AIC: $\dfrac{2}{n}(p+q)$ BIC: $\dfrac{\ln n}{n}(p+q)$

- Tests:
  1. Residual correlation:
     (a) Sample ACF: $r_k(\hat{\epsilon}) \sim N\left(0, \dfrac{1}{n}\right)$
     (b) Joint significance of first m ACF:
     $LB(m) = n(n+2)\sum \dfrac{r_k(\hat{\epsilon})^2}{n-k} \sim \chi^2_{m-p-q}$
  2. Normality: JB statistics
  3. Homoscedasticity: $Q(m) = n(n+2)\sum \dfrac{r_k(\hat{\epsilon}^2)^2}{n-k} \sim \chi^2_{m-p-q}$

Ch11. Spectral Analysis

- Spectral distribution function: $\gamma_h = \int_{-\pi}^{\pi} e^{i\pi v} dF(v)$
- Spectral density: $F(\lambda) = \int_{-\pi}^{\lambda} f(v)dv$
- ARMA spectral density: $f_y(\lambda) = \dfrac{\sigma^2}{2\pi}\dfrac{|\theta(e^{-i\pi})|^2}{|\phi(e^{-i\pi})|^2}$

Ch13. GARCH

- ARCH(q) : $\sigma_t^2 = \omega + \sum \alpha_i \epsilon_{t-i}^2$, $\epsilon_t = \sigma_t z_t$
  Weakly stationary iff $\sum \alpha_i < 1$
- GARCH(p,q) : $\sigma_t^2 = \omega + \sum \alpha_i \epsilon_{t-i}^2 + \sum \beta_j \sigma_{t-j}^2$, $\epsilon_t = \sigma_t z_t$
  Weakly stationary iff $\sum \alpha_i + \sum \beta_j < 1$
- GARCH(p,q) has unique s.s. solution iff $\gamma = E(\log(\beta + \alpha\eta_t^2)) < 0$

Ch14. Multivariate Time Series

- Weakly Stationary: $E(y_t) = \mu$, $E((y_t - \mu)(y_{t-k} - \mu)') = \Gamma_k$
- VAR(1) : $y_t = \phi_0 + \Phi y_{t-1} + \epsilon_t$
  Weakly stationary iff roots of $\det(I - \Phi z)$ outside unit circle
- VAR(n) : $y_t = \phi_0 + \sum \Phi_i y_{t-i} + \epsilon_t$
  Weakly stationary iff roots of $\det\left(I - \sum \Phi_i z^i\right)$ outside unit circle
- Invertibility: $y_t = \dfrac{\epsilon_t}{\Phi(B)}$
- Granger causality